



Tribhuvan University

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A Lab Report

On

Programming Logic & Techniques using Python (MCA504)

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Write a function that computes factorial of a number if the input number is more than 5.

```
def factorial(n):  
    if n > 5:  
        result = 1  
        for i in range(1, n + 1):  
            result *= i  
        return result  
  
print(factorial(6))  
print(factorial(5))
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>  
720  
None
```

Write a function that takes a number as input and checks whether the input number is prime or not.

```
def is_prime(n):  
    if n <= 1:  
        return False  
    for i in range(2, n, 1):  
        if n % i == 0:  
            return False  
    return True  
  
print(is_prime(7))  
print(is_prime(10))
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>  
True  
False
```

Write a function that take start and end number as inputs & display counting in your screen. (use time.sleep(sec) to take a pause while counting)

```
import time

def counting(start, end):
    for i in range(start, end + 1):
        print(i)
        time.sleep(1)

counting(1, 10)
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>
1
2
3
4
5
6
7
8
9
10
```

Write a function which accepts an argument and returns the data type of that argument.

```
def data_type(dt):
    return type(dt)
print(data_type(5))
print(data_type(3.14))
print(data_type("Hello"))
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>
<class 'int'>
<class 'float'>
<class 'str'>
```

Write a function that computes hypotenuse of a right-angled triangle by following the steps given below.

Hypotenuse² = Base² + Perpendicular²

a) Take base and perpendicular as inputs.

b) Create a function calculateSquare() for calculating and returning square of a number.

c) Create a function calculateHypotenuse() for calculating hypotenuse of a right angle triangle. Make use of the calculateSquare() function.

```
import math
def calculateSquare(num):
    return num ** 2

def calculateHypotenuse(base, perpendicular):
    hypotenuse_squared = calculateSquare(base) + calculateSquare(perpendicular)
    hypotenuse = math.sqrt(hypotenuse_squared)
    return hypotenuse

base = float(input("Enter the base of the triangle: "))
perpendicular = float(input("Enter the perpendicular of the triangle: "))

hypotenuse = calculateHypotenuse(base, perpendicular)
print(f"The hypotenuse of the triangle is: {hypotenuse}")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>
Enter the base of the triangle: 4
Enter the perpendicular of the triangle: 3
The hypotenuse of the triangle is: 5.0
```

Write a recursive function to compute the sum of numbers up to the input number.

```
def sum_up_to(n):
    if n <= 0:
        return 0
    else:
        return n + sum_up_to(n-1)

n = int(input("Enter a number: "))
print(sum_up_to(n))
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>
Enter a number: 10
55
```

Write functions to add, subtract, multiply and divide two numbers. Use these functions to calculate the value of

$x = a(b+c)(c-a)/(a+b-c)$ where $a=5$, $b=2$ and $c=4$.

```
def add(x,y):  
    return x+y
```

```
def subtract(x,y):  
    return x-y
```

```
def multiply(x,y):  
    return x*y
```

```
def divide(x,y):  
    if y == 0:  
        return "Cannot divide by zero"  
    else:  
        return x/y
```

```
a = 5  
b = 2  
c = 4
```

```
calculation = divide(multiply(multiply(a, add(b,c)), subtract(c,a)), subtract(add(a,b), c))  
print("The value of x is:", calculation)
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>p  
The value of x is: -10.0
```

Write a function to find the power of a number using lambda function.

```
power = lambda x, y: x ** y
```

```
number = int(input("Enter the base number: "))  
exponent = int(input("Enter the exponent: "))
```

```
result = power(number, exponent)  
print(f"{number} raised to the power of {exponent} is: {result}")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>  
Enter the base number: 5  
Enter the exponent: 2  
5 raised to the power of 2 is: 25
```

Write a recursive function to implement Tower of Hanoi problem.

```
def tower_hanoi(n, a,b,c):
    if n == 1:
        print("Move disk 1 from rod", a, "to rod", c)
        return
    tower_hanoi(n-1, a, c, b)
    print("Move disk", n, "from rod", a, "to rod", c)
    tower_hanoi(n-1, b, a, c)
    print("Move disk", n, "from rod", a, "to rod", c)

n = 2
tower_hanoi(n, 'A', 'B', 'C')
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>p
Move disk 1 from rod A to rod B
Move disk 2 from rod A to rod C
Move disk 1 from rod B to rod C
Move disk 2 from rod A to rod C
```

Perform slicing operations on a 2-dimensional list of size 3*3.

```
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]

# Slicing the rows
print("First row:", matrix[0])
print("Second row:", matrix[1])
print("Third row:", matrix[2])

# Slicing the columns
print("First column:", [row[0] for row in matrix])
print("Second column:", [row[1] for row in matrix])
print("Third column:", [row[2] for row in matrix])
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>p
First row: [1, 2, 3]
Second row: [4, 5, 6]
Third row: [7, 8, 9]
First column: [1, 4, 7]
Second column: [2, 5, 8]
Third column: [3, 6, 9]
```

Create a list with the ordered pair $\{(2,3),(1,4),(4,2),(5,6),(6,2)\}$ and a function that returns the sum of two numbers. Use list comprehension to get the sum of each ordered pairs.

```
ordered_pairs = [(2,3), (1,4), (4,2), (5,6), (6,2)]
```

```
def sum_of_numbers(x, y):  
    return x + y
```

```
sums = [sum_of_numbers(x, y) for x, y in ordered_pairs]  
print("The sums of the ordered pairs are:", sums)
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>  
The sums of the ordered pairs are: [5, 5, 6, 11, 8]
```

Create a file with name Sample.txt and write some contents into it. Also read the contents of the file. Also copy the contents of the file to another file with name Copy.txt.

```
with open("Sample.txt", "w") as file:  
    file.write("Hello Bishal!\nThis is a sample file.\nIt contains some text data.")
```

```
with open("Sample.txt", "r") as file:  
    contents = file.read()  
    print("Contents of Sample.txt:")  
    print(contents)
```

```
with open("Copy.txt", "w") as copy_file:  
    copy_file.write(contents)
```

```
print("Contents have been copied to Copy.txt")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First Semester\Python\Lab Task\Assignment 2>  
Contents of Sample.txt:  
Hello Bishal!  
This is a sample file.  
It contains some text data.  
Contents have been copied to Copy.txt
```

```
Copy.txt  X  Sample.txt  
Copy.txt  
You, 2 days ago | 1 author (You)  
1 Hello Bishal!  
2 This is a sample file.  
3 It contains some text data.
```